DeXposure-Claw: An Agentic System for DeFi Risk Monitoring and Decision Recommendation

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Abstract

Decentralized finance exposes supervisors to fast-moving, networked credit risks in which shocks can propagate across protocols before human operators can inspect raw on-chain data. We introduce DeXposure-Claw, a forecast-grounded agentic supervision system that routes LLM decisions through structured evidence rather than raw transactions. This addresses a problem in the existing LLM-agent approaches that are poorly suited to this setting because they can over-read weak or stale evidence and recommend high-stakes interventions without a regulator-aligned way to measure false alarms. The system uses DeXposure-FM, a graph time-series foundation model, to forecast future exposure networks, converts those forecasts into monitor alerts and stress-scenario evidence, and applies data-health and confidence gates before emitting auditable supervisory tickets with rationales. On five years of weekly DeFi exposure graphs, DeXposure-Claw improves ticket F1 over a conservative rules baseline from 0.0076 to 0.0233, while a cheaper Sonnet 4.6 decision model further raises F1 to 0.0288, eliminates measured false interventions, and directionally improves explanation quality, positioning the system as a recall–explanation Pareto option for human-in-the-loop DeFi supervision rather than a replacement for conservative rules.

1 Introduction

Decentralized finance (DeFi) creates fast-moving networks of token-mediated credit exposure among lending protocols, decentralized exchanges, stablecoins, bridges, and yield aggregators. Recent crises such as Terra/Luna, FTX, and SVB/USDC show that shocks can propagate across this network before supervisors can manually inspect raw on-chain data (Vidal-Tomás et al., 2023). This motivates decision-support systems that can forecast emerging exposure risk, identify affected protocols, and recommend timely supervisory responses.

A straightforward approach is to build an LLM agent that reasons directly over on-chain measurements. We argue that this is unsafe for high-stakes supervision: general-purpose LLM agents may produce plausible rationales while over-reading incomplete, stale, or weak evidence, thereby triggering unnecessary high-severity interventions. Existing systemic-risk evaluations also often rank protocols by fractional exposure changes (Bertomeu et al., 2024; Gonon et al., 2025; Li et al., 2025), which can overemphasize small protocols and fail to reflect regulator-relevant loss priorities.

We introduce *DeXposure-Claw*, a forecast-grounded agentic system for DeFi risk monitoring and decision recommendation. Rather than asking an LLM to reason over raw transactions, the system routes decisions through structured forecasted evidence. A graph time-series foundation model forecasts future exposure networks; deterministic monitors and stress scenarios convert these forecasts into alerts, attribution information, scenario losses, uncertainty estimates, and data-health scores; an LLM then emits ranked supervisory tickets with targets, severities, and evidence-linked rationales. Data-health and confidence gates restrict intervention-level recommendations, and every ticket is released with its evidence bundle, rationale, and gate states for auditability.

We evaluate the system with DeXposure-Bench, a six-axis harness covering forecast quality, warning behavior, uncertainty calibration, stress-scenario fidelity, ticket quality, and robustness. Its decision axis uses a regulator-aligned absolute-loss ground truth, allowing false interventions and supervisory relevance to be measured directly. Across five years of weekly DeFi exposure graphs and eight reference implementations, the full DeXposure-Claw stack improves ticket F1 from 0.0076 for a conservative persistence-rules baseline to 0.0233. Replacing the main Opus 4.7 decision model with Sonnet 4.6 yields the strongest operating point, raising F1 to 0.0288, reducing measured false interventions to zero, and directionally improving explanation quality at lower inference cost. These results position DeXposure-Claw as an auditable recall–explanation Pareto option for human-in-the-loop DeFi supervision, rather than a replacement for conservative rule-based systems.

2 Related Work

Bench positioning and ground-truth definition. LLM-agent benchmarks (HELM (Liang et al., 2022), SWE-bench (Jimenez et al., 2024), AgentBench (Liu et al., 2024)) score open-ended reasoning, software repair, and generic agent behaviour; temporal-graph benchmarks (TGB (Huang et al., 2023), OGB (Hu et al., 2020)) score structural prediction quality. Neither scores whether an LLM agent's supervisory decisions match what a regulator would actually prioritise. The closest prior systemic-risk evaluations (Bertomeu et al., 2024; Gonon et al., 2025; Li et al., 2025) rank protocols by *fractional* weight change, which disproportionately surfaces tiny protocols of low systemic relevance. DeXposure-Bench replaces this with an *absolute-loss* ground truth (§3).

LLM agents in finance and DeFi. General agent patterns such as ReAct (Yao et al., 2023) combine reasoning with tool use; FinGPT (Yang et al., 2023) adapts language models to financial data. Within DeFi, agentic systems have so far targeted transaction auditing (Yao et al., 2026), intent mining (Mao et al., 2025), smart-contract verification (Hu et al., 2026; Kong et al., 2026), price manipulation (Liu et al., 2025), and explanation for graph anomalies (Watson et al., 2025); portfolio-oriented agents combine graph encoders with LLMs (Luo et al., 2025; Jeon and Lee, 2026). These systems consume raw transactions or token text, not structured forecasted evidence, and none reports a false-intervention rate against a regulator-aligned ground truth—a gap our characterisation (§5.1) closes.

Forecast-grounded LLM agents in other domains. Pairing a domain forecaster with an LLM decision layer is an emerging deployment pattern. In macroeconomics, ChatGPT-augmented PMI nowcasting (de Bondt and Sun, 2025) and LLM-driven macroeconomic forecasting (Carriero et al., 2025) both feed structured numeric inputs into an LLM that produces interpretable narratives; a BIS primer surveys the wider pattern (Kwon et al., 2024). Time-series foundation models such as Chronos (Ansari et al., 2024), Lag-Llama (Rasul et al., 2024), and TimesFM (Das et al., 2024) make the forecaster reusable across tasks; tabular foundation models extend the same idea to heterogeneous structured data (Hollmann et al., 2025; Eremeev et al., 2025). None of these forecast→LLM pipelines, to our knowledge, has been scored against a regulator-aligned ground truth for high-stakes financial-network supervision.

3 Preliminaries

We model decentralized-finance credit exposure as a temporal weighted directed graph. Let $\mathcal{P}$ be the universe of protocols and $\mathcal{X}$ the universe of tokens. At time t, protocol p holds tokens $X_p(t) \subseteq \mathcal{X}$, while protocol q issues or is economically linked to tokens $X_G(q) \subseteq \mathcal{X}$. Protocol p is exposed to q if

$$X_p(t) \cap X_G(q) \neq \emptyset. \tag{1}$$

The exposure network is

$$G_t = (V_t, E_t, W_t), \tag{2}$$

where $V_t \subseteq \mathcal{P}$ is the active protocol set, $E_t \subseteq V_t \times V_t$ is the directed exposure set, and $W_t[p,q] = w_{pq,t} \geq 0$ is the USD exposure from p to q. For protocol v, define its incident exposure mass as

$$w_t(v) = \sum_{u \in V_t} w_{uv,t} + \sum_{u \in V_t} w_{vu,t}. \tag{3}$$

The observed history is a sequence of snapshots $\mathcal{G}_{1:T} = \{G_1, \ldots, G_T\}$, optionally with covariates X_t.

At decision epoch t, the forecasting problem is to estimate G_{t+h} for horizons $h \in \mathcal{H}$. A probabilistic temporal-graph forecaster outputs

$$P_{t,h} = P(G_{t+h} \mid G_{\leq t}, X_{\leq t}). \tag{4}$$

For each ordered pair (p,q), let

$$\pi_{pq,t+h} = \Pr(e_{pq,t+h} = 1 \mid G_{\leq t}, X_{\leq t}), \tag{5}$$

and

$$\mu_{pq,t+h} = \mathbb{E}[w_{pq,t+h} \mid e_{pq,t+h} = 1, G_{\leq t}, X_{\leq t}]\,. \tag{6}$$

A representative predicted graph $\widehat{G}_{t+h}$ is constructed by

$$\widehat{E}_{t+h} = \{(p,q) : \pi_{pq,t+h} > \pi_{\min}\},$$
$$\widehat{w}_{pq,t+h} = \pi_{pq,t+h}\mu_{pq,t+h}.$$

Uncertainty is represented by Monte Carlo samples

$$\widetilde{G}^{(1)}_{t+h}, \ldots, \widetilde{G}^{(M)}_{t+h} \sim P_{t,h}. \tag{7}$$

A monitor maps graphs to scalar risk statistics $\Phi_j : \mathcal{G} \to \mathbb{R}$, such as maximum PageRank, HHI concentration, density, PageRank Gini, or degree Gini. Given a rolling window of length L, define

$$\mu_{j,t} = \frac{1}{L}\sum_{\ell=1}^{L} \Phi_j(G_{t-\ell}),$$

$$\sigma_{j,t} = \sqrt{\frac{1}{L-1}\sum_{\ell=1}^{L}(\Phi_j(G_{t-\ell}) - \mu_{j,t})^2}.$$

An alert fires when

$$A_{j,t,h} = \mathbf{1}\left\{\frac{\left|\Phi_j(\widehat{G}_{t+h}) - \mu_{j,t}\right|}{\max(\sigma_{j,t}, \epsilon)} > z\right\}. \tag{8}$$

A stress scenario is a graph perturbation $s : \mathcal{G} \to \mathcal{G}$. Let G^s be the shocked graph and define

$$\mathrm{Mass}(G) = \sum_{e \in E(G)} w_e,$$

$$L(s; G) = \frac{\mathrm{Mass}(G) - \mathrm{Mass}(G^s)}{\mathrm{Mass}(G)}.$$

For sampled graphs, let $\ell_m(s,h) = L(s; \widetilde{G}^{(m)}_{t+h})$. If $\ell_{(1)} \geq \cdots \geq \ell_{(M)}$ are sorted losses, then

$$\mathrm{CVaR}_\lambda(s,h) = \frac{1}{\lceil \lambda M \rceil}\sum_{m=1}^{\lceil \lambda M \rceil} \ell_{(m)}. \tag{9}$$

For evaluation, we define regulator-aligned ground truth by absolute exposure loss. For protocol v and horizon h,

$$\Delta^h_t(v) = w_t(v) - w_{t+h}(v). \tag{10}$$

Given percentile $\pi \in (0,1)$, the stressed set is

$$S^h_t = \mathrm{top}_\pi \left\{ v : w_t(v) > 0,\ \Delta^h_t(v) > 0 \right\}, \tag{11}$$

where protocols are ranked by $\Delta^h_t(v)$.

A supervisory system emits tickets

$$D_t = \{\tau_{t,1}, \ldots, \tau_{t,K_t}\}, \qquad \tau_{t,k} = (u_{t,k}, Y_{t,k}, r_{t,k}), \tag{12}$$

where $u_{t,k}$ is an action, $Y_{t,k} \subseteq V_t$ is a target set, and $r_{t,k}$ is a rationale. The action space is

$$\mathcal{U} = \{\textsc{Monitor}, \textsc{Investigate}, \textsc{RecommendReduce}, \textsc{Contingency}\}.$$

For predicted targets $\widehat{S}_t$, ticket quality is

$$\mathrm{Precision} = \frac{|\widehat{S}_t \cap S^h_t|}{|\widehat{S}_t|},$$

$$\mathrm{Recall} = \frac{|\widehat{S}_t \cap S^h_t|}{|S^h_t|},$$

$$\mathrm{F1} = \frac{2\ \mathrm{Precision}\ \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}.$$

Let $\widehat{S}^{\mathrm{int}}_t$ be targets of intervention-level actions in $\{\textsc{RecommendReduce}, \textsc{Contingency}\}$. The false-intervention rate is

$$\mathrm{FIR} = \frac{|\widehat{S}^{\mathrm{int}}_t \setminus S^h_t|}{|\widehat{S}^{\mathrm{int}}_t|}. \tag{13}$$

Finally, define the data-health score

$$\mathrm{DH}_t = \frac{1}{4}\left(\mathrm{fresh}_t + \mathrm{miss}_t + \mathrm{topo}_t + \mathrm{disc}_t\right). \tag{14}$$

For alert a at horizon h, with Monte Carlo coefficient of variation $\mathrm{CV}_{t,h}(a)$, define

$$C_t(a) = \mathrm{DH}_t \cdot \frac{1}{1 + \mathrm{CV}_{t,h}(a)} \cdot \frac{1}{1 + \log(1+h)}. \tag{15}$$

If $\bar{C}_t$ is the mean active-alert confidence, intervention-level actions are feasible only when

$$\mathrm{DH}_t \geq \tau_{\mathrm{data}} \quad \text{and} \quad \bar{C}_t \geq \tau_{\mathrm{conf}}. \tag{16}$$

4 DeXposure-Claw Pipeline

This section introduces the DeXposure-Claw pipeline.

4.1 Overview

DeXposure-Claw is a four-layer pipeline for forecast-grounded DeFi supervision (Figure 1). Layer 1 uses DeXposure-FM to forecast near-future credit-exposure graphs. Layer 2 converts the forecast distribution into a typed evidence bundle containing monitor alerts, uncertainty estimates, and stress-scenario losses. Layer 3 passes this bundle to an LLM, which emits ranked supervisory tickets with targets, severities, and rationales. Layer 4 applies data-health and confidence gates to determine whether intervention-level actions are admissible. The pipeline runs at weekly cadence on a single RTX 4090, with well under a dollar of LLM API spend per weekly decision (a full DeXposure-Bench leaderboard run is approximately $10).

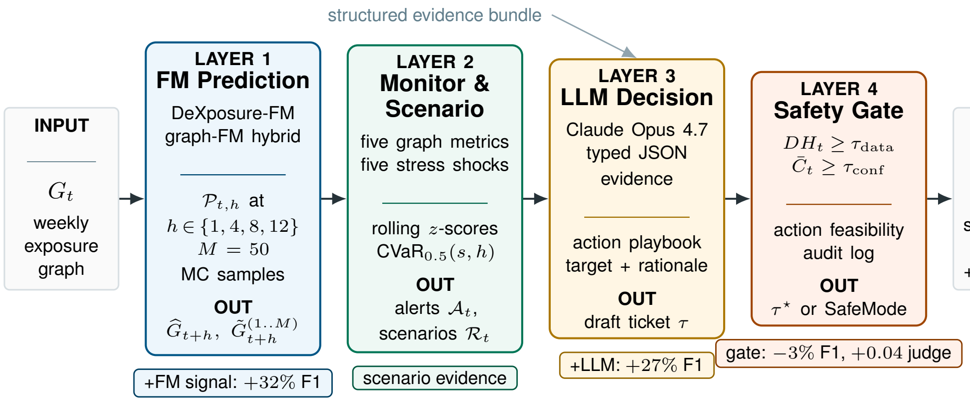

Figure 1: DeXposure-Claw: a four-layer supervisory pipeline. Given a weekly exposure graph G_t, the system first forecasts future exposure networks with a foundation model (Layer 1), converts those forecasts into monitor alerts and stress-scenario evidence (Layer 2), asks an LLM to produce supervisory tickets from the typed evidence bundle (Layer 3), and applies safety gates before releasing either a ticket $\tau^\star$ or a safe-mode notice (Layer 4). Bottom annotations report the per-layer F1 and judge-score lifts from Section 5.1.

4.2 Forecaster interface

At each decision epoch t, the agent observes a credit-exposure snapshot $G_t = (V_t, E_t, W_t)$, an inter-protocol token-mediated exposure graph defined in Section 3, together with node covariates X_t. For each forecast horizon $h \in \mathcal{H} = \{1, 4, 8, 12\}$ weeks, it queries DeXposure-FM for a predictive distribution

$$\mathcal{P}_{t,h} \leftarrow \mathrm{ForecastFM}(G_t, X_t, h). \qquad (17)$$

From this distribution, the agent constructs an expected-weight predicted graph $\widehat{G}_{t+h}$ by thresholding edge-existence probabilities at $\pi_{\min} = 0.2$. It also draws $M = 50$ Monte Carlo samples to quantify predictive uncertainty. The predicted-graph construction is defined in Section 3; implementation choices are given in Appendix A.1.

4.3 Structured evidence

Layer 2 converts $\widehat{G}_{t+h}$ and its Monte Carlo samples into a typed evidence bundle. Five *monitor functionals* summarize concentration, fragility, and instability in the predicted graph: top-1 PageRank (Brin and Page, 1998), HHI concentration (Rhoades, 1993), network density, PageRank Gini, and degree Gini. An alert fires when a monitor statistic deviates from its $L = 42$ week rolling baseline by more than $z = 2$ standard deviations. Each alert includes two additional fields: (i) an attribution top-K list of contributing edges, ranked by marginal contribution, and (ii) an uncertainty-aware confidence score $C_t(a) \in [0, 1]$ that decreases with lower data health, wider Monte Carlo dispersion, and longer forecast horizons.

In parallel, a *scenario engine* applies five standardized stress shocks to both $\widehat{G}_{t+h}$ and each Monte Carlo sample: single protocol failure, bridge cluster failure, stablecoin de-peg, sector lending shock, and correlated top-10 stress. For each scenario s and horizon h, the engine aggregates the worst-half sampled losses as $\mathrm{CVaR}_{0.5}(s, h)$. Full monitor and scenario definitions are provided in Appendix A.

4.4 LLM decision layer

Layer 3 receives the evidence bundle as a typed JSON payload containing alerts with attribution top-K, a scenario-CVaR table, Monte Carlo dispersion summaries, and the scalar data-health score DH_t. The LLM is prompted to emit a ranked list of supervisory tickets. Each ticket contains (a) a severity level from a fixed four-action playbook (`Monitor`, `Investigate`, `Recommend-Reduce`, and `Contingency`), (b) a target set of protocols, and (c) a free-form rationale that cites specific evidence fields by name. To measure self-consistency, we repeat the decision call three times at temperature 0 and report the Jaccard overlap of ticket targets and severities.

4.5 Safety gates

Layer 4 determines whether intervention-level recommendations are feasible. The *data-health gate* computes

$$\mathrm{DH}_t = \mathrm{mean}\left(\mathrm{fresh}_t, \mathrm{miss}_t, \mathrm{topo}_t, \mathrm{disc}_t\right). \quad (18)$$

If $\mathrm{DH}_t < \tau_{\mathrm{data}} = 0.7$, the system enters safe mode and emits only `Monitor` or `Investigate` tickets, regardless of the LLM's recommended severity. The *confidence gate* blocks intervention-level tickets whenever the mean alert confidence satisfies $\bar{C}_t < \tau_{\text{conf}} = 0.6$, even if the data-health gate passes. Every emitted ticket carries the full evidence bundle, the LLM rationale, and the gate states that produced it. The resulting audit log is the unit of release (Section 6).

The end-to-end procedure is given in Appendix A (Algorithm A.1).

5 Experiments

Axes and ground truth. DeXposure-Bench scores the pipeline on six axes: forecast quality (`b1_forecast`), warning lead time (`b2_warning`), uncertainty calibration (`b3_calibration`), stress-scenario losses (`b4_stress`), regulator-prioritised tickets (`b5_decision`), and degraded-data robustness (`b6_robustness`). The decision and warning axes share the absolute-loss ground truth of §3, avoiding fractional-change rankings that overweight tiny protocols. We fix $\pi = 0.05$ and $h = 4$ weeks, yielding $|\mathcal{S}_t^h| \approx 283$ protocols per week; decision axes score $n = 29$ test weeks because the final four snapshots lack a 4-week-ahead target.

Data, splits, and references. We use DeXposure weekly exposure graphs (4,300 protocols, 24,300 tokens, 43.7M exposure entries, 283 snapshots from 2020-03 to 2025-08) (Wu et al., 2025). All model selection uses pre-2025 data: 2020-03–2024-06 train and 2024-07–2024-12 validation; 2025-01–2025-08 is the frozen test split. The harness bundles eight reference implementations: `h1`, `m1`–`m7`, spanning heuristic monitors, persistence, snapshot LLM, EvolveGCN (Pareja et al., 2020), DeXposure-FM, rules, LLM decisions, and safety gates. Full schemas and method details are in §B.2–§B.3; the historical Terra/Luna, FTX, and SVB/USDC warning study is reported separately from the 2025 leaderboard.

Reproducibility. A complete leaderboard run finishes in under four GPU-hours on a single RTX 4090 plus approximately $10 of LLM API spend. The harness fixes seeds, versions, and checkpoint hashes; every reported number reproduces from the released artefacts.

5.1 Results

We use DeXposure-Bench to ask three questions. Does DeXposure-FM provide forecast signal that persistence does not (`b1`, `b3`, `b6`)? Does routing the LLM through structured evidence improve supervisory tickets once precision, recall, explanation quality, and false interventions are scored jointly (`b4`, `b5`)? Which components are load-bearing, and which are reserves under degradation (ablations)?

DeXposure-FM contributes trend, calibration, and robustness; persistence still wins on static point error. On static error metrics, persistence is a deceptively strong baseline: at $h = 4$ it beats the FM on PageRank MAE (3.4×10^{-5} vs 4.5×10^{-5}) and edges it on Spearman rank correlation (0.570 vs 0.558). The FM contributes signal where persistence is structurally pinned: trend consistency 0.525–0.632 across horizons (persistence: 0); `b3_calibration` PI coverage 0.913 against a 0.90 target and ECE 0.013; `b6_robustness` mean relative degradation 0.113 vs 0.148 for persistence (24% less). The full per-horizon and per-regime audits are in §C.

Headline decision-quality result. Table 1 consolidates the central `b5_decision` comparison. The persistence+rules baseline (`m1`) achieves the highest single-axis precision (0.720); like `m2`/`m5` it never escalates beyond `Investigate`, so its FIR is undefined, not a safety guarantee. Against this baseline, the full DeXposure-FM+LLM+gate stack (`m7`) lifts ticket F1 from 0.0076 to 0.0233 ($+207\%$; $p < 10^{-4}$). Against the comparable pure-LLM baseline that consumes raw snapshots without the forecaster (`m2`, the only other LLM-bearing row), `m7` lifts LLM-as-judge explanation quality from 2.24 to 2.45 on a 1–5 scale ($+9.4\%$) under a tier-above Claude Opus 4.8 judge (Zheng et al., 2023); the lift is significant under GPT-5.5 ($p < 0.001$) but not Opus 4.8 ($p = 0.23$; §C.2)—directional rather than confirmed. The cost surfaces sharply once `m2` is in the table: the raw-snapshot LLM issues *no* interventions, but adding the FM signal (`m2`→`m6`) makes it escalate—and 44.8% of those interventions misfire (FIR $0 \to 0.448$; $p < 10^{-4}$). The safety gate (`m6`→`m7`) reduces FIR only marginally ($0.448 \to 0.437$), and its effects on F1 and judge score are within noise (§C.2). Both LLM variants reach grounding 1.000, so the failure mode is not citation hallucination but *over-reading the forecaster's evidence as warrant for high-severity action.*[1]

[1] `m3_evolvegcn` and `m4_fm_only` are predictor-only and do not emit tickets (forecast quality in §C.1); EvolveGCN trails persistence downstream, consistent with its ∼220

Judge robustness across model families. We re-judge m2/m6/m7 with a cross-family panel (Table 2). All three frontier judges rank m7 highest (mean score varying by ≤ 0.45). The one disagreement—Gemini 2.5 Pro scores m2 above m6 (2.69 vs 2.55), still below m7—sits within Gemini's $\sim 3\times$ higher weekly noise (std 1.3–1.5 vs 0.3–0.6). Cross-family agreement on the top system closes the judge-family confound.

A tier-below, cheaper decision LLM dominates the headline operating point. We re-run the decision layer of m6 and m7 on the *same* FM-derived prompts with two non-Opus 4.7 decision LLMs—Claude Sonnet 4.6 (within-family, tier-below, $\sim 5\times$ cheaper) and Google Gemini 2.5 Pro (cross-family)—and re-score with the same Opus 4.8 judge (Table 3). On both architectures Sonnet 4.6 is a Pareto improvement over Opus 4.7: F1 rises (+15%/+24%; $p < 0.001$ on m7), judge quality is directionally higher (+0.11/+0.21; n.s.), and FIR collapses from 0.437–0.448 to 0.000—genuinely: Sonnet 4.6 still issues interventions at Opus-comparable volume ($\sim$190 vs 252), all on truly-stressed targets. Gemini 2.5 Pro is more conservative still (lower F1, FIR 0), so over-intervention is a property of the decision model, not of LLM decision layers per se. The operating point a practitioner would now deploy is m7 with Sonnet 4.6 (F1 0.0288, FIR 0.000, judge 2.66).

Layer-wise contribution. Holding the decision policy fixed at *LLM* and adding the DeXposure-FM signal lifts ticket F1 by 32% ($0.0183 \to 0.0241$; $p < 0.001$); holding the predictor fixed at DeXposure-FM and swapping rules for the LLM lifts F1 by another 27% ($0.0190 \to 0.0241$). The two layers therefore make additive, non-substitutable contributions (Figure C.1 in §C). Adding the safety gate trades 3% F1 for judge changes within noise; its value is operational—every intervention clears two gates with auditable logs.

Matched-budget view. F1 conflates target quality with volume: m6/m7 emit 6.3–6.5 targets per week vs 4.9 for m2 and 2.1 for m1. Truncating every method to its top-k targets per week ($k \in \{1, 2, 3\}$; §C.2) separates the two effects: the LLM variants still roughly double the rules baseline's precision@k ($p \leq 0.0055$), but the FM-fed and raw-snapshot LLM variants become statistically indistinguishable. The FM's contribution is thus not a sharper top of the ranking but support for a *larger* target set at undiminished precision; under Opus 4.7 the expanded tail is where over-intervention concentrates.

Ablation: load-bearing vs reserve components. Under clean 2025 data the confidence gate is load-bearing: removing it raises FIR from 0.000 to 0.429. The scenario engine is load-bearing for target extraction: skipping it collapses ticket precision from 0.600 to 0.000. The data-health gate and multi-horizon diversity are dormant on clean data but activate as reserves under degradation. Under severe feature/edge masking (80–98%) the data-health gate suppresses up to 60% of interventions that would otherwise produce false alarms (FIR range $[0.27, 0.60]$ when disabled vs 0.000 when triggered); multi-horizon monitoring generates $\sim 4\times$ more alerts in the three crisis windows at unchanged precision. Per-regime ablation audits are in §C.1.

Early-warning timeline (historical event study). For the three pre-test crisis windows the shared weighted-degree monitor (h1) achieves median lead times of 4–5 weeks with precision 1.000 at SVB/USDC across all alert budgets $K \in \{5, 10, 20\}$; per-budget details in §C.1. This is historical monitor evidence, not a test-split result.

6 Discussion and Conclusion

The experiments point to three findings that a single-axis leaderboard would miss. First, the FM's value is not a sharper point forecast but trend, calibration, and robustness upstream and coverage downstream. Persistence is a strong point-error baseline and matched-budget precision makes FM-fed and raw-snapshot LLM variants indistinguishable; yet where persistence is structurally pinned the FM supplies trend consistency (0.525–0.632 vs 0), calibrated intervals (PI coverage 0.913, ECE 0.013), and 24% lower degradation under stress, and downstream its evidence supports more targets at undiminished precision and lifts F1. Second, the LLM failure mode is actionability calibration, not grounding. Opus 4.7 cites valid fields (grounding 1.000) but has FIR = 0.437–0.448; Sonnet 4.6 on the same FM prompts raises F1 to 0.0288 and reduces FIR to 0.000. FIR is therefore a first-class metric alongside grounding and judge scores. Third, gates are reserve capacity: clean-data effects are within noise, but under severe masking the data-

weekly-snapshot training budget.

ID	Pred.	Decision	Prec.$\uparrow$	Recall$\uparrow$	F1$\uparrow$	Judge$\uparrow$	FIR$\downarrow$	Ground.$\uparrow$	Stabil.$\uparrow$
`m1`	persist.	rules	**0.720**	0.004	0.0076	—	—	—	0.514
`m2`	—	LLM (no FM)	0.575	0.009	0.0183	2.24	—	1.000	0.532
`m5`	DeXposure-FM	rules	0.600	0.010	0.0190	—	—	—	0.257
`m6`	DeXposure-FM	LLM	0.570	0.012	**0.0241**	2.41	0.448	1.000	0.488
`m7`	DeXposure-FM	LLM + safety gate	0.580	0.012	0.0233	**2.45**	0.437	1.000	0.435

Table 1: Headline `b5_decision` comparison on the frozen 2025 test split ($n = 29$). Rows are read as a multi-objective trade-off rather than a single leaderboard: `m1` is the conservative high-precision baseline; `m2` is the pure-LLM baseline (no forecaster); `m6`/`m7` are the FM-grounded LLM variants. Judge scores (1–5 scale) are from a Claude Opus 4.8 judge, strictly tier-above the Claude Opus 4.7 decision LLM; cross-family agreement is in Table 2. — in FIR: undefined— `m1`/`m2`/`m5` emit no intervention-level ticket that could misfire.

Method	Opus 4.8	Gemini 2.5 Pro	GPT-5.5	Haiku 4.5[†]
`m2`	2.24	2.69	1.90	2.03
`m6`	2.41	2.55	2.45	2.48
`m7`	**2.45**	**2.90**	2.45	2.69

Table 2: Cross-family judge panel on the same Opus 4.7 decisions ($n = 29$). Gemini and GPT use $\mathrm{reasoning} = \{\mathrm{effort} : \mathrm{minimal}\}$ so the thinking budget matches the Anthropic baseline. [†] Haiku 4.5 is a tier-below weak-judge reference.

Method	Decision LLM	F1$\uparrow$	FIR$\downarrow$	Judge$\uparrow$
`m6`	Opus 4.7 (main)	0.0241	0.448	2.41
`m6`	Sonnet 4.6	**0.0276**	**0.000**	**2.52**
`m6`	Gemini 2.5 Pro	0.0129	0.000	2.21
`m7`	Opus 4.7 (main)	0.0233	0.437	2.45
`m7`	Sonnet 4.6	**0.0288**	**0.000**	**2.66**
`m7`	Gemini 2.5 Pro	0.0139	0.000	2.38

Table 3: Cross-decision-LLM panel on the same FM-derived prompts, judge fixed at Claude Opus 4.8 ($n = 29$). Sonnet 4.6 dominates Opus 4.7 on both architectures. The bold row (`m7` $\times$ Sonnet 4.6) beats every row in Table 1 on every LLM-bearing axis at $\sim 5\times$ lower cost.

health gate suppresses up to 60% of FIR-eligible tickets, so gates need stress tests and audit logs that support threshold retuning.

Overall, DeXposure-Claw is best read as an auditable recall–explanation Pareto option for human-in-the-loop DeFi supervision: structured forecast evidence improves coverage and rationales, but high-severity recommendations still require decision-model calibration and intervention-realism measurement.

Limitations

Historical replay evaluation. We evaluate DeXposure-Claw by replaying 283 frozen weekly snapshots of DeXposure (2020–2025), which lets us compare methods under identical evidence and audit calibration, lead time, and ticket quality. This setup does not measure production-integration factors such as live data-feed reliability, operator workload, or threshold retuning under changing supervisory priorities. These factors affect deployment policy rather than the central comparison studied here: whether structured forecast evidence improves auditable human-in-the-loop risk monitoring.

Single domain. DeXposure-Bench evaluates a single on-chain risk surface (DeFi inter-protocol credit exposure). Generalisation to NFT lending, perpetual derivatives, or cross-chain bridge networks is untested and will require domain-specific re-calibration of the conformal split and the stressed-set percentile π.

Weekly resolution. The pipeline operates at weekly granularity, but several DeFi shocks unfold within hours—Terra/Luna erased \$40B in under 48 hours. A sub-hourly production deployment would require denser snapshot ingestion and re-evaluation of conformal calibration on the resulting denser time series. The 4–5 week lead times we report should be read as "how early the weekly monitor turns", not as end-to-end alert latency.

Judge thinking-budget is held minimal. The cross-family judge panel (Claude Opus 4.8, Google Gemini 2.5 Pro, OpenAI GPT-5.5) is run with reasoning effort set to *minimal* so the comparison against the non-reasoning Anthropic baseline is a like-for-like comparison of intrinsic model capability rather than thinking budget. Default reasoning mode on Gemini 2.5 Pro and GPT-5.5 would change the reported scores; we treat the reasoning-on configuration as a distinct evaluation surface and leave it for a follow-up study. Practitioners deploying the judge at default reasoning settings should re-calibrate the score thresholds reported here. Human-judge calibration with practising supervisors remains the most pressing methodological extension.

Decision-outcome proxy. `b5_decision` scores tickets against a regulator-aligned absolute-loss ground truth, which captures supervisory targeting priority but not the counterfactual effect of supervisory action—whether the recommended risk reduction would have changed the subsequent contagion path. We therefore interpret decision scores as evidence of targeting and explanation quality, not as a causal estimate of intervention impact.

Ethics Statement

Data provenance. DeXposure is built from public on-chain measurements aggregated via DefiLlama (DefiLlama, 2026). No private user data, wallet-level personally-identifying information, or off-chain identity data is used. Protocol-level aggregates are the unit of analysis throughout.

Decision risk. DeXposure-Claw emits supervisory tickets, not automated trades or automated interventions. The high-severity action types (`Recommend-Reduce`, `Contingency`) are gated on a data-health score and a confidence score, and the pipeline emits a safe-mode notice when either gate fails. We characterise the system as decision support for a human supervisor, not as an autonomous risk agent. The measured false-intervention rate (FIR ≈ 0.44 for the DeXposure-FM+LLM variant) is reported in the headline results table so that downstream operators can weigh the trade-off against their own risk tolerance before deployment.

LLM use disclosure. The decision-layer LLM (`m2`, `m6`, `m7`) is Claude Opus 4.7 in the main configuration; Claude Sonnet 4.6 and Google Gemini 2.5 Pro are reported as decision-layer ablations. The LLM-as-judge for the explanation-quality axis is a three-model cross-family panel: Claude Opus 4.8 (within-family upgrade, strictly tier-above Opus 4.7), Google Gemini 2.5 Pro, and OpenAI GPT-5.5 (cross-family). All judges are run with reasoning effort set to *minimal* so the comparison is controlled for thinking budget. Claude Haiku 4.5 from a prior run is reported in the appendix as a weak-judge baseline. Every model is accessed through the OpenRouter unified API at temperature 0.

Release. We will release the harness, the eight reference implementations, and the ticket-level audit logs under a permissive open licence. Model checkpoints will be released subject to dataset-licence compatibility checks.

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A Additional Algorithm Details

Section 3 defines the exposure graph, forecast distribution, predicted graph, monitor alert rule, scenario loss, ticket metrics, and gate feasibility conditions used throughout the paper. This appendix records only the concrete instantiation used by DeXposure-Claw.

A.1 Forecast instantiation

At each decision epoch t, DeXposure-Claw queries DeXposure-FM once for each horizon $h \in \{1, 4, 8, 12\}$ and constructs $\widehat{G}_{t+h}$ with the expected-weight operator from Section 3. We set the edge-existence threshold to $\pi_{\min} = 0.2$, tuned on the validation split. Uncertainty summaries use $M = 50$ Monte Carlo graph samples. Each sample first draws edge indicators from the predicted edge probabilities, then perturbs positive edge weights with the forecaster's predictive regression spread. For existing edges, residual predictions are folded into the conditional mean $\mu_{pq,t+h}$ before applying the expected-weight construction; candidate new edges enter only when their predicted existence probability exceeds $\pi_{\min}$.

A.2 Monitor instantiation

ID	Metric	Definition
N1	Systemic Importance	$\max_{p \in V} \mathrm{PageRank}(\widehat{G}_{t+h})_p$
N2	HHI Concentration	$\sum_p (d_p / \sum_q d_q)^2$, $d_p = \sum_q \widehat{w}_{pq,t+h}$
N3	Network Density	$\lvert\widehat{E}_{t+h}\rvert / \lvert V\rvert(\lvert V\rvert - 1)$
N4	PageRank Gini	$\mathrm{Gini}(\{\mathrm{PageRank}_p\}_{p \in V})$
N5	Degree Gini	$\mathrm{Gini}(\{d_p\}_{p \in V})$

All monitors use the rolling alert rule from Section 3 with a 42-week baseline, $z = 2$, and $\epsilon = 10^{-8}$. PageRank uses damping 0.85 and 10 power iterations. Gini coefficients are computed over active nodes in the predicted graph.

A.3 Scenario instantiation

ID	Scenario	Target	Shock
S1	Single protocol fail	Highest-weight node	100%
S2	Bridge cluster fail	All bridge nodes	100%
S3	Stablecoin de-peg	All stablecoin nodes	50%
S4	Sector lending shock	All lending nodes	30%
S5	Correlated stress	Top-10 by weight	20%

Scenario targets use protocol-sector metadata from DeXposure. For each scenario and horizon, DeXposure-Claw reports the deterministic loss on $\widehat{G}_{t+h}$ and the sampled $\mathrm{CVaR}_{0.5}$ defined in Section 3.

A.4 Gate instantiation and ticket scoring

The data-health gate uses the four checks in Section 3 with $\tau_{\mathrm{data}} = 0.7$. Safe mode still releases monitor alerts and scenario summaries, but suppresses intervention-level tickets (RECOMMENDREDUCE and CONTINGENCY). The confidence gate uses $\tau_{\mathrm{conf}} = 0.6$ and clamps each Monte Carlo coefficient of variation to $[0, 1]$ before computing alert confidence.

Feasible tickets are ranked by

$$\mathrm{Score}_t(u) = w_u \cdot \bar{C}_t \cdot \mathrm{Imp}_t, \tag{19}$$

where $w_u \in \{0.25, 0.50, 0.75, 1.00\}$ is the playbook severity weight for the four ordered actions and Imp_t is the mean scenario CVaR (default 1.0 when no scenario summary is available).

A.5 End-to-end procedure

B Additional Implementation Details

B.1 Forecaster architecture and training

DeXposure-FM is built on the GraphPFN graph-tabular foundation backbone (Hollmann et al., 2025; Eremeev et al., 2025) with a LiMiX-16M tabular transformer encoder (12 layers, 4 attention heads, hidden dimension 64). The same backbone serves three heads—edge existence (binary), edge weight (regression),

Algorithm A.1 DeXposure-Claw: one decision epoch.

Require: (G_t, X_t); horizons $\mathcal{H}$; scenarios $\mathcal{S}_0$; thresholds $(\tau_{\mathrm{data}}, \tau_{\mathrm{conf}}, z, \pi_{\min})$; MC count M
Ensure: Alerts $\mathcal{A}_t$; scenario summary $\mathcal{R}_t$; tickets $\mathcal{D}_t$

$\mathrm{DH}_t \leftarrow \mathrm{DataHealth}(G_t, X_t)$
$\mathrm{SAFE\_MODE}_t \leftarrow \mathbb{I}\{\mathrm{DH}_t < \tau_{\mathrm{data}}\}$
for all $h \in \mathcal{H}$ **do**
  $P_{t,h} \leftarrow$ DeXposure-FM.Forecast(G_t, X_t, h)
  $\widehat{G}_{t+h} \leftarrow \mathrm{BuildPredGraph}(P_{t,h}; \pi_{\min})$
  $\{\widetilde{G}^{(m)}_{t+h}\}_{m=1}^{M} \leftarrow \mathrm{MCSample}(P_{t,h})$
  Compute monitor statistics on $\widehat{G}_{t+h}$, attach rolling baselines, and derive alerts $\mathcal{A}_{t,h}$
  Compute alert confidences $C_t(\cdot)$ from sampled monitor dispersion
  for all $s \in \mathcal{S}_0$ **do**
    Compute deterministic loss $L(s; \widehat{G}_{t+h})$ and sampled $\mathrm{CVaR}_{0.5}(s, h)$
  end for
end for
$\mathcal{D}_t \leftarrow \mathrm{PlaybookDecision}(\mathcal{A}_t, \mathcal{R}_t, \mathrm{DH}_t, \mathrm{SAFE\_MODE}_t, \tau_{\mathrm{conf}})$
return $(\mathcal{A}_t, \mathcal{R}_t, \mathcal{D}_t)$

and node-TVL change (regression)—trained jointly with a BCE+MSE composite loss. We fine-tune one checkpoint per horizon $h \in \{1, 4, 8, 12\}$ for 20 epochs with AdamW (learning rate 10^{-4} for heads, 10^{-5} for the backbone). Fine-tuning draws on DeXposure snapshots from March 2020 to January 2025 (104 training weeks, 12 validation weeks for early stopping, 8 held-out weeks for internal evaluation); all data that influences weights or checkpoint selection precedes the frozen 2025 leaderboard split (§B.2). Full-graph in-context inference fits a single RTX 4090, and one weekly forecast across all four horizons completes in well under a minute. These intrinsic forecaster numbers are kept out of the main text because the paper's contribution is the operating point of the full pipeline, not the FM's headline accuracy.

B.2 Benchmark details

Six-axis schema.

ID	Capability tested	Primary metrics
`b1_forecast`	Temporal graph prediction	PageRank/HHI/Gini MAE, trend consistency, Spearman; per $h \in \{1, 4, 8, 12\}$
`b2_warning`	Streaming anomaly detection	Precision, recall, F1, lead time, alert stability; per event $\times$ budget
`b3_calibration`	Predictive uncertainty	PI coverage at 0.90 target, PI width, ECE, CRPS
`b4_stress`	What-if scenario fidelity	Loss MAE, distressed-count MAE, propagation depth MAE, target overlap@k
`b5_decision`	Supervisory ticket quality	Precision, recall@k, F1, target stability, judge score, FIR
`b6_robustness`	Data quality sensitivity	Per-benchmark metrics under 5 degradation regimes; relative degradation

The six axes are non-overlapping (a system can excel at `b1` while failing `b5`), decomposable along the four-layer framework so that predictor (`b1`, `b3`, `b6`), monitor (`b2`), scenario (`b4`), and decision (`b5`) ablations report independently, and individually publishable so downstream work may extend a single axis without re-running the suite.

Dataset split.

Split	Period	Weeks	Usage
Train	2020-03–2024-06	$\sim$222	Forecaster training
Validation	2024-07–2024-12	$\sim$26	Conformal calibration, threshold tuning
Test	2025-01–2025-08	$\sim$33	Frozen leaderboard

Historical warning windows. The `b2_warning` event study is separate from the frozen 2025 leaderboard split. It evaluates the shared weighted-degree monitor around Terra/Luna (2022-05-09), FTX (2022-11-07), and SVB/USDC (2023-03-10) with a shorter 26-week rolling baseline, chosen so each event window has enough pre-event history. The 2025 leaderboard runs use the 42-week monitor baseline in Appendix A.

Ground-truth definition. With $w_t(v) = \sum_{e \ni v} w_t(e)$ the total weight incident on v, the regulator-aligned stressed set at horizon h is

$$\Delta_t^h(v) = w_t(v) - w_{t+h}(v), \quad \mathcal{S}_t^h = \mathrm{top}_\pi\{v : w_t(v) > 0,\ \Delta_t^h(v) > 0\}, \tag{20}$$

with $\pi = 0.05$ and ground-truth horizon $h = 4$ weeks. The final four test weeks lack the 4-week-ahead snapshot w_{t+4} and are not scored, leaving $n = 29$ evaluated weeks. This is the single ground-truth definition used by `b2`, `b5`, and the LLM-decision pipeline. A submission produces one JSON file per {benchmark, method} pair plus a single `results.json`; the harness fixes random seeds, snapshots library versions, and emits SHA-256 hashes of all checkpoint files.

B.3 Reference methods

ID	Predictor	Policy	Role
`h1_weighted_degree`	—	heuristic alert	Streaming anomaly baseline (`b2`)
`m1_persistence_rules`	$G_{t+h} = G_t$	rule engine	Decision baseline
`m2_snapshot_llm`	current snap	LLM	LLM-only ablation (no forecast)
`m3_evolvegcn`	EvolveGCN	—	Learned-GNN baseline
`m4_fm_only`	DeXposure-FM	—	FM-only ablation
`m5_fm_rules`	DeXposure-FM	rule engine	FM + rules
`m6_fm_llm`	DeXposure-FM	LLM	Full FM + LLM stack
`m7_fm_llm_gated`	DeXposure-FM	LLM + safety gate	Safety-gated FM + LLM

These methods expose all four cuts of the four-layer ablation: adding the predictor (`m1`→`m5`), swapping the decision layer (`m5`→`m6`), adding the safety gate (`m6`→`m7`), and removing the predictor from the LLM stack (`m2`↔`m6`). All conform to a shared `predict_graph`/`decide_actions` interface.

LLM model panel and decoding. All LLM-bearing methods (`m2`, `m6`, `m7`) are evaluated under three decision-layer configurations: Claude Opus 4.7 (main), Claude Sonnet 4.6 (within-family, tier-below ablation), and Google Gemini 2.5 Pro (cross-family ablation). Each runs at temperature 0 and $\max\_tokens = 4096$; self-consistency is probed with three repeated calls per week and Jaccard overlap on emitted ticket targets and severities. The LLM-as-judge for the explanation-quality axis is a three-model panel chosen so the judge tier is at or above the decision tier: Claude Opus 4.8 (strictly tier-above Opus 4.7), Google Gemini 2.5 Pro, and OpenAI GPT-5.5; the Gemini and GPT judges run with OpenRouter's $\mathrm{reasoning} = \{\mathrm{effort} : \mathrm{minimal}\}$ so the judge thinking budget matches the non-reasoning Anthropic baseline. Claude Haiku 4.5 is reported as a weak-judge reference. All models are served through OpenRouter; observed cost is on the order of a few US dollars per full sweep of the 29-week test split.

C Additional Experimental Results

This appendix expands the headline numbers of §5.1 into the per-slice audits underlying them, then reports week-level uncertainty for the central comparisons.

C.1 Per-slice result tables

Full all-horizon `b1_forecast` results.

Per-scenario `b4_stress` results.

Per-regime `b6_robustness` results.

Per-budget `b2_warning` results.

Per-regime A1 (data-health gate) isolation.

Stress-condition ablation Panels B and C. Panel B: under severe feature/edge masking, FIR ranges over $[0.27, 0.60]$ when the data-health gate is disabled ($\tau_{\mathrm{data}} = 0$; the confidence gate is held off at $\tau_{\mathrm{conf}} = 0$ throughout Panel B to isolate A1), and collapses to 0.000 when the gate is triggered, by suppressing intervention tickets. Panel C: multi-horizon monitoring ($h \in \{1, 4, 8, 12\}$) generates 73–98 alerts in the three crisis windows vs 19–23 for single-horizon ($h = 4$), at unchanged precision range 0.533–0.624 and zero false interventions in both cases.

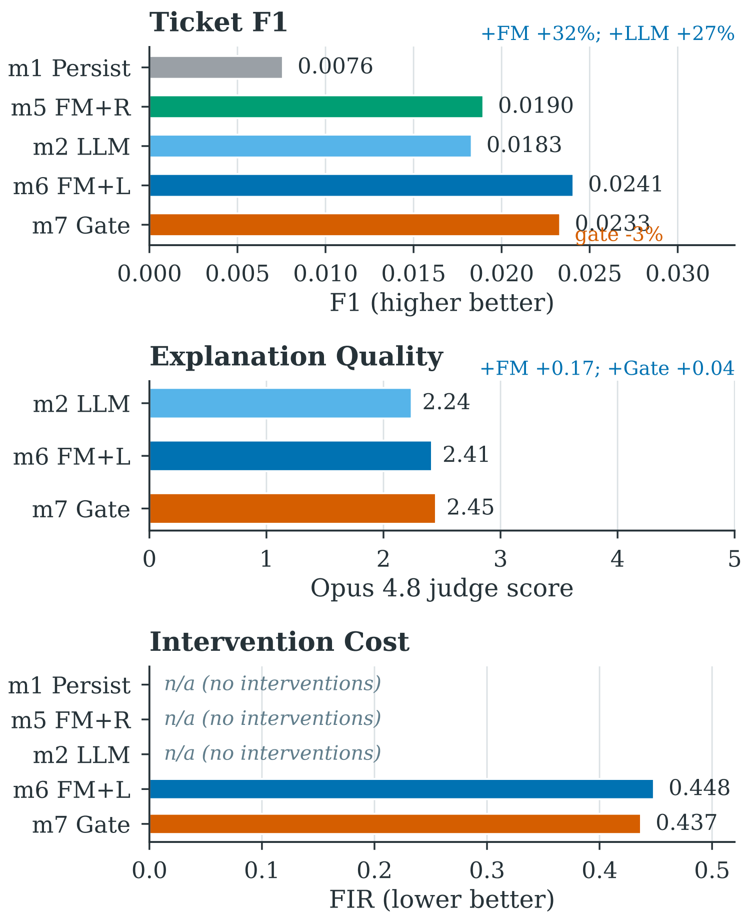

Figure C.1: Layer-wise contribution and cost (visualising Tables 1 and 3). Adding the FM signal to a snapshot LLM improves ticket F1 and explanation quality; swapping the FM-fed rule engine for the LLM improves recall-oriented F1 further; the same LLM variants incur a substantial false-intervention rate, partially mitigated by the safety gate. FIR is marked *n/a* for `m1`/`m5`/`m2` (as "—" in Table 1): these methods emit no intervention-level ticket, so FIR is undefined $(0/0)$ rather than a genuine zero.

Table C.1: `b1_forecast` all-horizon forecasting on the 2025 test split. PageRank MAE shown in 10^{-5} units. Best per horizon and metric in **bold**; ties share the bold. Trend consistency for `m1_persistence` is structurally 0 because $\hat{G}_{t+h} = G_t$ produces no trend signal.

h	Method	PR MAE (↓)	HHI MAE (↓)	Gini MAE (↓)	Rank Corr (↑)	Trend Cons (↑)
1	`m1_persistence`	**3.6**	**0.0376**	**0.0027**	**0.556**	0.000
	`m3_evolvegcn`	3.7	0.0481	0.1149	0.535	0.444
	`m4_fm_only`	4.7	**0.0376**	0.0029	0.549	**0.531**
	`m5_fm_rules`	4.7	**0.0376**	0.0029	0.549	0.525
4	`m1_persistence`	**3.4**	0.0322	**0.0018**	**0.570**	0.000
	`m3_evolvegcn`	3.5	0.0489	0.1326	0.551	0.324
	`m4_fm_only`	4.5	**0.0321**	0.0022	0.559	0.621
	`m5_fm_rules`	4.5	**0.0321**	0.0022	0.558	**0.628**
8	`m1_persistence`	**3.9**	0.0399	**0.0028**	**0.517**	0.000
	`m3_evolvegcn`	4.0	0.0492	0.1140	0.499	0.296
	`m4_fm_only`	4.5	**0.0398**	0.0030	0.508	0.624
	`m5_fm_rules`	4.5	**0.0398**	0.0030	0.508	**0.632**
12	`m1_persistence`	**3.7**	0.0393	**0.0027**	**0.525**	0.000
	`m3_evolvegcn`	3.8	0.0534	0.1245	0.509	0.314
	`m4_fm_only`	4.1	**0.0392**	0.0029	0.516	**0.600**
	`m5_fm_rules`	4.1	**0.0392**	0.0029	0.516	**0.600**

Table C.2: `b4_stress` per-scenario detail. Loss MAE in normalised units; Distress count MAE in protocols; target overlap@10 measured against the actual future graph. Best per scenario and metric in **bold**; ties share the bold, and columns where all four methods tie are left unbolded. The main pattern is conservative: FM variants roughly match the strong persistence baseline on loss and target-overlap metrics, while EvolveGCN is generally weaker on downstream what-if fidelity.

Scenario	Method	Loss MAE (↓)	Distress MAE (↓)	Prop. Depth MAE (↓)	Overlap@10 (↑)
S1 (single protocol)	`m1_persistence`	0.0668	**200.3**	**0.897**	**0.314**
	`m3_evolvegcn`	0.1072	218.1	0.966	0.034
	`m4_fm_only`	**0.0667**	201.4	**0.897**	**0.314**
	`m5_fm_rules`	**0.0667**	201.6	**0.897**	**0.314**
S2 (bridge cluster)	`m1_persistence`	0.0150	**27.3**	0.000	0.521
	`m3_evolvegcn`	**0.0116**	196.4	0.000	0.189
	`m4_fm_only`	0.0151	50.3	0.000	**0.525**
	`m5_fm_rules`	0.0151	52.1	0.000	**0.525**
S3 (stablecoin de-peg)	`m1_persistence`	**0.0079**	0.0	0.000	**0.445**
	`m3_evolvegcn`	0.0156	0.0	0.000	0.269
	`m4_fm_only`	0.0080	0.0	0.000	**0.445**
	`m5_fm_rules`	0.0080	0.0	0.000	**0.445**
S4 (sector lending)	`m1_persistence`	0.0100	0.0	0.034	**0.536**
	`m3_evolvegcn`	0.0200	0.0	0.034	0.292
	`m4_fm_only`	**0.0099**	0.0	0.034	**0.536**
	`m5_fm_rules`	**0.0099**	0.0	0.034	**0.536**
S5 (correlated top-10)	`m1_persistence`	0.0100	0.0	**0.000**	**0.602**
	`m3_evolvegcn`	0.0538	0.0	1.000	0.330
	`m4_fm_only`	**0.0099**	0.0	**0.000**	**0.602**
	`m5_fm_rules`	**0.0099**	0.0	**0.000**	**0.602**

Table C.3: `b6_robustness` per-regime detail. All metrics at h=4 on the 2025 test split under five degradation regimes. PageRank, HHI, and Gini MAE in 10^{-3} units. Relative degradation is signed against the clean `b1_forecast` baseline; negative values mean the predictor is no worse than on clean data. Best per metric (lowest MAE, highest rank correlation, smallest $|\Delta_{\text{rel}}|$) in **bold**; ties share the bold. Persistence ignores node features, so its `noisy_features_01` and `missing_features_20` rows degenerate to the clean numbers (and $\Delta_{\text{rel}} = 0$); EvolveGCN's predictions also change only negligibly under these two regimes ($|\Delta_{\text{rel}}| < 10^{-8}$).

Regime	Method	PR MAE (↓)	HHI MAE (↓)	Gini MAE (↓)	Rank Corr (↑)	Δ_{rel} (~0)
low_data_10pct	`m1_persistence`	15.94	0.0591	**0.0023**	**0.552**	$+0.766$
	`m3_evolvegcn`	15.23	0.0999	0.1301	0.535	$\mathbf{+0.276}$
	`m4_fm_only`	12.83	**0.0586**	0.0027	0.528	$+0.612$
	`m5_fm_rules`	**12.71**	0.0587	0.0027	0.528	$+0.607$
low_data_25pct	`m1_persistence`	**9.46**	0.0282	**0.0020**	**0.577**	-0.093
	`m3_evolvegcn`	10.22	0.0486	0.1281	0.558	$\mathbf{-0.027}$
	`m4_fm_only`	9.86	**0.0279**	0.0024	0.553	-0.126
	`m5_fm_rules`	9.95	**0.0279**	0.0024	0.553	-0.126
partial_graph_30	`m1_persistence`	**9.63**	**0.0350**	**0.0019**	**0.554**	$+0.068$
	`m3_evolvegcn`	10.36	0.0488	0.1236	0.537	$\mathbf{-0.048}$
	`m4_fm_only`	12.52	**0.0350**	0.0022	0.500	$+0.082$
	`m5_fm_rules`	12.45	**0.0350**	0.0022	0.499	$+0.077$
noisy_features_01	`m1_persistence`	**9.77**	**0.0322**	**0.0018**	**0.584**	$\mathbf{0.000}$
	`m3_evolvegcn`	10.68	0.0489	0.1326	0.563	$\mathbf{0.000}$
	`m4_fm_only`	11.38	0.0323	0.0021	0.554	-0.005
	`m5_fm_rules`	11.35	0.0323	0.0021	0.554	-0.007
missing_features_20	`m1_persistence`	**9.77**	**0.0322**	**0.0018**	**0.584**	$\mathbf{0.000}$
	`m3_evolvegcn`	10.68	0.0489	0.1326	0.563	$\mathbf{0.000}$
	`m4_fm_only`	12.11	0.0323	0.0023	0.559	$+0.017$
	`m5_fm_rules`	12.08	0.0323	0.0023	0.559	$+0.014$

Table C.4: `b2_warning` per-event detail for the shared weighted-degree monitor (`h1_weighted_degree`) across alert budgets $K \in \{5, 10, 20\}$. Recall and F1-warning shown in 10^{-3} units (the full DeFi corpus contains ~4,300 protocols, so per-week recall denominators are large). Lead time is event-level and not budget-sensitive: 5 weeks for Terra/Luna and FTX, 4 weeks for SVB-USDC. `b2_warning` sits upstream of the predictor choice, so a single monitor is shared across all methods (see §5.1).

Event	K	Prec. (↑)	Rec. $\times 10^3$ (↑)	$\mathrm{F1_w}$ $\times 10^3$ (↑)	Stab. (↑)	$\lvert\mathcal{A}\rvert$
Terra/Luna	5	0.60	1.81	3.61	0.44	11
	10	0.70	3.26	6.49	0.62	19
	20	0.65	7.97	15.74	0.74	36
FTX	5	0.60	1.33	2.65	0.64	10
	10	0.70	2.39	4.77	0.72	18
	20	0.55	4.79	9.49	0.72	39
SVB-USDC	5	1.00	1.81	3.62	0.55	10
	10	1.00	2.82	5.62	0.75	15
	20	1.00	5.64	11.22	0.81	30

Table C.5: Per-regime data-health gate ablation isolating A1 from A2. Each row holds τ_{conf}=0 and sweeps τ_{data}. "*strict*" = τ_{data}=0.85, "*on*" = 0.70, "*off*" = 0.00. The four regimes apply targeted edge or feature degradation on the clean 2025 test split (`n_weeks` = 29): `extreme` masks both topology and features at 90%, `topo` removes 90% of edges only, `feat` corrupts 90% of node features only, `severe` masks both at 80%. Numbers expand Panel B of Table C.6: under degradation the gate-off setting produces FIR up to 0.541, while *on*/*strict* can suppress intervention tickets via safe-mode and drive FIR to 0.

Regime	Gate	τ_{data}	$\overline{\mathrm{dh}}$	Safe-mode (%)	Ticket Prec. (↑)	FIR (↓)	Interventions
extreme	strict	0.85	0.587	100	0.510	**0.000**	0
	on	0.70	0.587	100	0.545	**0.000**	0
	off	0.00	0.587	0	0.490	0.510	57
extreme_topo	strict	0.85	0.637	100	0.517	**0.000**	0
	on	0.70	0.637	100	0.462	**0.000**	0
	off	0.00	0.637	0	0.559	0.441	58
extreme_feat	strict	0.85	0.755	100	0.503	**0.000**	0
	on	0.70	0.755	0	0.510	0.267	5
	off	0.00	0.755	0	0.503	0.267	5
severe	strict	0.85	0.751	100	0.490	**0.000**	0
	on	0.70	0.751	0	0.538	0.400	25
	off	0.00	0.751	0	0.538	0.541	24

Table C.6: Component-level ablation of DeXposure-Claw on `b5_decision`. Panel A shows clean 2025 test-split ablations against the `m5_fm_rules` baseline; Panels B–C report targeted stress runs for mechanisms that are dormant on clean data. A2 (confidence gating) and A3 (scenario engine) are load-bearing in the clean setting, while A1 (data-health gating) and A6 (multi-horizon monitoring) are safety-reserve mechanisms whose effects appear under data degradation and crisis-window dynamics, respectively.

Panel	Ablation	Configuration	Ticket Prec.	FIR	Notes
A: clean test	—	full `m5_fm_rules`	0.600	0.000	baseline
	A1	disable data-health gate (τ_{data}=0)	0.600	0.000	no effect (clean data)
	A2	disable confidence gate (τ_{conf}=0)	0.600	0.429	FIR shoots up
	A3	skip scenario engine	0.000	0.000	target extraction collapses
	A6	single-horizon forecasting (h=4)	0.600	0.000	no effect (clean data)
B: degraded data	A1 off	80–98% feature/edge mask, τ_{conf}=0	—	0.27–0.60	false interventions emerge
	A1 on	same degradation, safe-mode triggered	—	0.000	intervention suppression
C: crisis period	A6: single	horizons = $\{4\}$, crisis windows	0.533–0.624	0.000	alerts: 19–23
	A6: multi	horizons = $\{1, 4, 8, 12\}$, same windows	0.533–0.624	0.000	alerts: 73–98 ($\approx 4\times$)

C.2 Statistical significance and matched-budget analysis

All headline metrics are means over the $n = 29$ scored test weeks. We report 95% percentile bootstrap confidence intervals (10,000 resamples of weeks with replacement; F1 recomputed from the resampled mean precision and recall) and two-sided paired sign-flip permutation tests (20,000 permutations, methods aligned by week). Analysis code is released with the harness (`scripts/bootstrap_stats.py`, `scripts/matched_budget.py`).

Table C.7: 95% bootstrap CIs for the decision-quality headline numbers (decision LLM in parentheses; judge is Claude Opus 4.8 throughout). ‡`m1`'s per-week distribution comes from a local re-run of the released harness: its aggregate F1 (0.0081) brackets the published GPU-run value (0.0076), and the recall@k and stressed-pool size reproduce exactly, but with <1 ticket per week a few borderline tickets flip across environments (re-run precision 0.633 vs published 0.720), so the `m1` interval should be read as order-of-magnitude.

Configuration	F1 [95% CI]	FIR [95% CI]	Judge [95% CI]
`m1` rules‡	0.0081 [0.0041, 0.0123]	0.000	—
`m2` (Opus 4.7)	0.0184 [0.0142, 0.0223]	0.000	2.24 [2.10, 2.41]
`m6` (Opus 4.7)	0.0241 [0.0201, 0.0281]	0.448 [0.293, 0.601]	2.41 [2.21, 2.62]
`m7` (Opus 4.7)	0.0234 [0.0194, 0.0271]	0.437 [0.282, 0.592]	2.45 [2.24, 2.66]
`m6` (Sonnet 4.6)	0.0276 [0.0225, 0.0324]	0.000	2.52 [2.28, 2.79]
`m7` (Sonnet 4.6)	0.0288 [0.0236, 0.0335]	0.000	2.66 [2.41, 2.90]
`m6` (Gemini 2.5 Pro)	0.0129 [0.0103, 0.0155]	0.000	2.21
`m7` (Gemini 2.5 Pro)	0.0139 [0.0111, 0.0167]	0.000	2.38

Table C.8: Two-sided paired permutation p-values for the key comparisons. Significant effects ($p < 0.05$) in **bold**.

Comparison	Metric	p
`m6` vs `m2` (FM signal)	F1	**0.0002**
`m7` vs `m1` (full stack)	F1	**<0.0001**
`m7` vs `m6` (gate)	F1	0.41
`m7`×Sonnet vs `m7`×Opus	F1	**0.0004**
`m6` vs `m2`	FIR	**<0.0001**
`m6` vs `m2` (Opus 4.8 judge)	Judge	0.23
`m6` vs `m2` (GPT-5.5 judge)	Judge	**0.0002**
`m6` vs `m2` (Gemini judge)	Judge	0.77
`m7` vs `m6` (Opus 4.8 judge)	Judge	1.00
`m7`×Sonnet vs `m7`×Opus	Judge	0.21

The load-bearing quantitative claims—the FM signal's F1 lift, the full stack's F1 lift over rules, the Sonnet 4.6 F1 improvement, and the over-intervention cost—are significant at $p < 0.001$. The explanation-quality lift of the FM-grounded variants is judge-dependent (significant under GPT-5.5, positive but not significant under Opus 4.8 at $p = 0.23$, absent under the noisiest Gemini judge), so we read the judge axis as directional evidence. The safety gate's effects on F1 and judge score are indistinguishable from noise on clean data, consistent with its characterisation as a reserve mechanism (§6).

Matched-budget recall@k. Methods emit different numbers of ticket targets per week (`m1` 2.1, `m2` 4.9, `m7` 6.3, `m6` 6.5, `m7`$\times$Sonnet 7.6), and unbudgeted recall mechanically favours methods that flag more protocols. To separate target *quality* from target *volume*, we truncate every method to its top-k targets per week (LLM variants ranked by their own emitted risk score; `m1` by ticket confidence) and score against the same stressed pool.

Table C.9: Matched-budget recall@k ($\times 10^3$) and precision@k on the 2025 test split. At matched budgets every LLM variant beats the rules baseline ($p \leq 0.0055$ vs `m1` at all k), while the FM-grounded and raw-snapshot LLM variants are statistically indistinguishable from each other (all pairwise $p \geq 0.10$). `m1` row from the local re-run (Table C.7 note).

	$k=1$		$k=2$		$k=3$	
Method	R@1	P@1	R@2	P@2	R@3	P@3
`m1` rules	0.71	0.24	1.71	0.28	2.45	0.26
`m2` (Opus 4.7)	1.53	0.52	3.47	0.55	5.70	0.59
`m6` (Opus 4.7)	1.64	0.55	3.29	0.53	5.30	0.56
`m7` (Opus 4.7)	1.64	0.55	3.47	0.55	5.30	0.56
`m7` (Sonnet 4.6)	1.64	0.55	3.47	0.55	5.30	0.56

The matched-budget view decomposes the headline F1 result. Against the rules baseline, LLM targeting is genuinely better per target: at every k, precision@k roughly doubles (0.52–0.59 vs 0.24–0.28) and recall@k is significantly higher ($p \leq 0.0055$). Against the raw-snapshot LLM, the FM's contribution is *not* a higher per-target hit rate—at matched k the two are indistinguishable—but the ability to support a *larger* target set at undiminished precision (Table 1). That expanded tail is also where the over-intervention risk concentrates: under Opus 4.7 it is escalated to intervention-level severity ($\mathrm{FIR} = 0.448$), while Sonnet 4.6 expands the same set without escalating it ($\mathrm{FIR} = 0$).